Scenario

You have been asked to design a data pipeline for an AI system that predicts customer demand for an online store. The system will use both sales data and social media mentions to generate accurate demand forecasts.

To complete this activity, you will outline how data flows through the following stages:

1. **Data Ingestion** – How data enters your system from different sources.
2. **Data Processing and Transformation** – How you clean, combine, and prepare data.
3. **Data Storage** – Where and how the processed data will be stored for analysis or training.
4. **Model Integration** – How the AI model will access the processed data for prediction.

Step 1: Identify Data Sources

Begin by listing the sources your AI system will use. Consider both structured and unstructured data.

Examples:

- Structured: CRM sales data, order histories, product inventories.
- Unstructured: Social media posts, customer reviews, or website click logs. </aside>

Describe how each data source will connect to the pipeline. For instance, will it come through an API, a database connection, or a web scraping tool?

Step 2: Define the Data Processing Workflow

Next, describe how raw data will be cleaned, formatted, and transformed. Think about how the ETL (Extract, Transform, Load) process applies to your use case.

Consider:

- How to handle missing or duplicate data.
- How to combine structured and unstructured sources.
- Whether you will normalise or scale numeric values.
- Which tools or methods you might use, such as Power BI, AWS Glue, or Google Dataflow.

Step 3: Select a Storage Solution

Decide where you will store your processed data.

Questions to guide your choice:

- Do you need a relational or NoSQL database?
- Will you store your data locally or in the cloud?
- How will you ensure the storage system is scalable and secure? </aside>

Possible options include:

- **Relational:** MySQL, PostgreSQL, or Azure SQL Database.
- **NoSQL:** MongoDB, Cassandra, or Firebase.
- **Cloud Storage:** AWS S3, Google Cloud Storage, or Microsoft Azure Blob Storage.

Step 4: Visualise Your Pipeline

Create a **flowchart or diagram** that represents how data moves through your pipeline.

You can use tool such as:

- Draw.io
- Lucidchart
- Miro
- Canva

Your diagram should clearly show the following components:

1. Data sources (input)
2. Processing and transformation stage
3. Storage and access layer
4. Connection to the AI model

Label each part and include brief notes on what happens in each step.

Step 5: Add Security and Compliance Considerations

Briefly outline how you will protect the data moving through your pipeline.

You might include:

- Encryption of data at rest and in transit.
- User access control policies.
- Regular data audits for compliance with GDPR or POPIA. <

This step ensures that your pipeline not only performs efficiently but also meets privacy and ethical standards.

Step 6: Reflect on Design Choices

Once your flowchart is complete, reflect on your decisions. Consider the following questions:

- What trade-offs did you make between speed, cost, and data quality?
- How would you modify your pipeline to scale for larger datasets?
- Which parts of your pipeline could be automated to improve efficiency?

Please ensure that your assignment is uploaded in PDF format. The diagram may also be submitted as a PDF.

Grading

Marks for each of the criteria will be assigned accordingly:

- 0 - Failed to submit
- 1 - Limited
- 2 - Basic
- 3 - Proficient
- 4 - Advanced
- 5 - Exceptional

The following assessment criteria will be used to grade your submission:

- Effort
- Application
- Examples
- AI for Business insight

Rubric

Criteria	**1**	**2**	**3**	**4**	**5**
	Limited	**Basic**	**Proficient**	**Advanced**	**Exceptional**
Effort	Minimal effort apparent, with little attention to detail or engagement with the task.	Shows some effort, but responses may lack polish or thoroughness.	Demonstrates a solid effort, with responses showing attention to detail and engagement with the task.	Displays a strong effort, with responses exhibiting thoroughness, coherence, and depth.	Exemplifies an outstanding effort, with responses displaying exceptional clarity, insight, and completeness.

Application	Fails to apply learned information effectively to the scenario.	Demonstrates basic application of learned information to the scenario, but lacks depth or coherence.	Applies learned information to the scenario with clarity and coherence, demonstrating a solid understanding of its relevance.	Applies learned information skillfully to the scenario, offering insightful analysis and connections.	Applies learned information in a highly sophisticated manner, integrating multiple perspectives and drawing nuanced conclusions.
Examples	Few or no examples provided. Demonstrates limited understanding or application of concepts.	Provides some examples, but they may be simplistic or lack depth. Demonstrates basic understanding of concepts.	Offers several relevant examples demonstrating understanding of concepts. Examples are well-integrated into the response.	Presents numerous detailed examples that vividly illustrate concepts and their applications.	Presents numerous detailed examples that vividly illustrate concepts and their applications.
AI for Business Insight	Shows little understanding of the principles of AI for Business Strategy and fails to connect theories to the scenario.	Displays basic understanding of AI for Business Strategy; makes simple, expected connections to the scenario.	Demonstrates a clear understanding of AI for Business Strategy and makes relevant connections to the scenario, reflecting solid insight.	Exhibits deep understanding of AI for Business Strategy concepts; offers thoughtful, original insights that enhance the scenario analysis.	Demonstrates an understanding of AI for Business while; integrating sophisticated insights that significantly enrich the scenario, reflecting a high level of critical thinking.